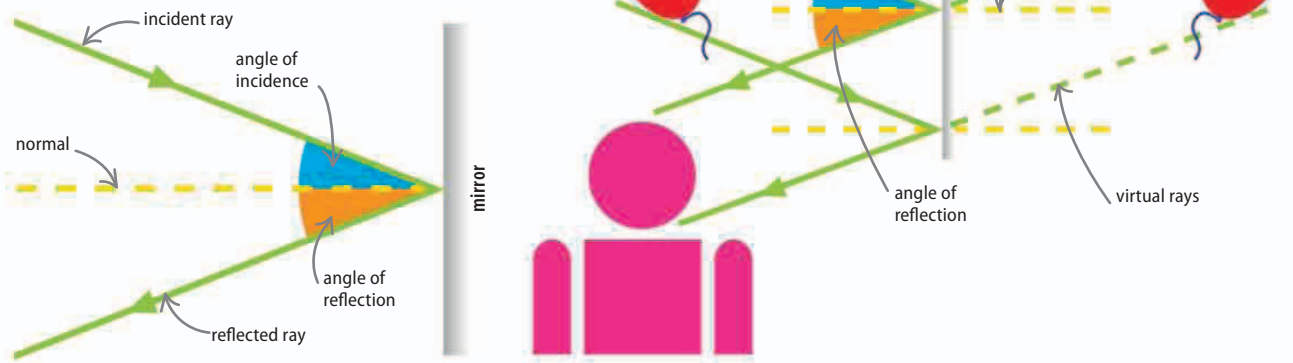


Reflection

When rays hit a smooth, shiny, flat surface, such as a flat mirror, they are reflected perfectly to give a clear but reversed image. Rough surfaces cause light to bounce off in different directions, so there is no reflected image.



△ Angles of incidence and reflection

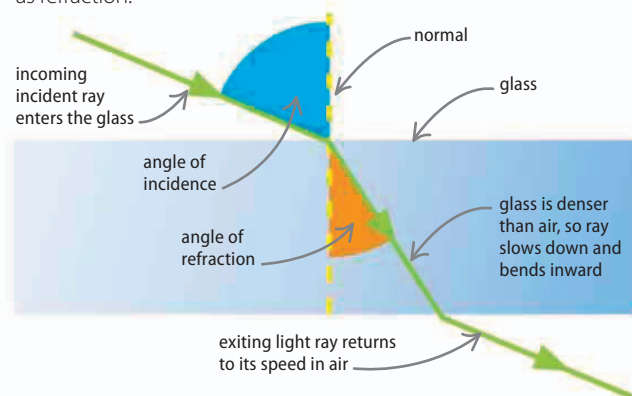
A reflection is made up of an incoming ray, called the incident ray, and an outgoing ray, called the reflected ray. The angle of incidence is equal to the angle of reflection, measured from a imaginary line at 90° to the mirror, called the normal.

△ Virtual image

The image in a mirror appears to be behind it—light rays appear to be focused there, but they do not actually meet at that point. This is called a virtual image. The image on a movie screen is called a real image because rays from the projector focus directly on the screen.

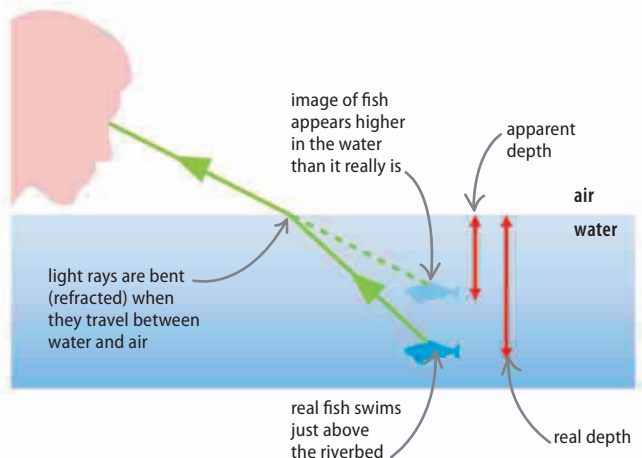
Refraction

Light rays usually travel in straight lines, but pass through different media (materials)—such as air, water, or glass—at different speeds. When light moves from one medium to another, the change in speed makes the beam change direction. This effect is known as refraction.



△ Changing direction

If light travels through air and then enters at an angle a more dense medium, such as glass, the rays slow down and refract inward. They travel in a straight line through the glass but at an angle to their original direction. As the rays pass out from the glass to the air, they return to their original path and speed up again.



△ Real and apparent depth

Light rays refract when they pass from water to the lighter medium of air. This means that when you look from an angle at an object in water, it is not in fact where you see it. A fish swimming in the water is actually deeper than it appears to be.